EART27201— Sedimentary Rocks and Fossils
Fossil Practicals — Microfossils

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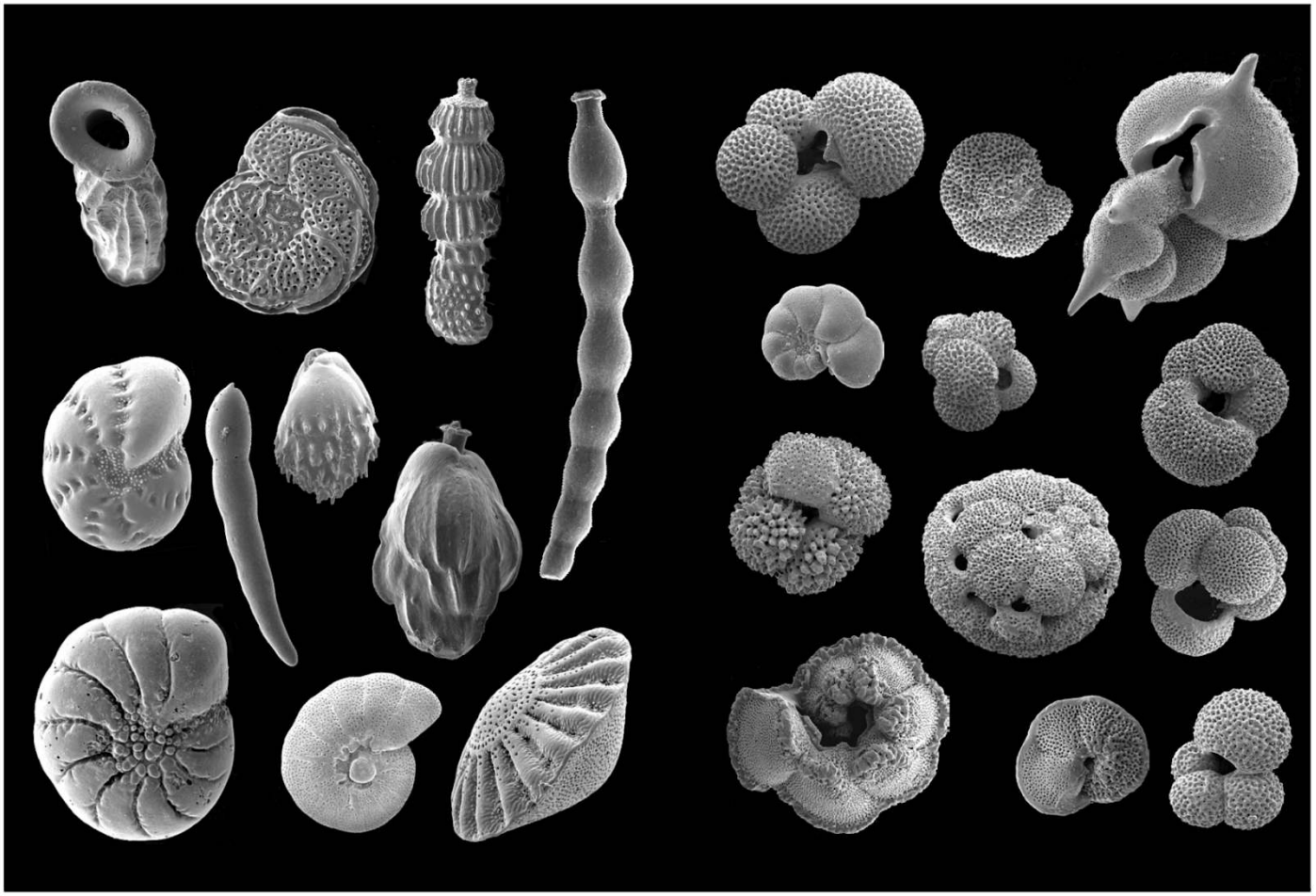
Aim: This practical will give you hands-on experience of using a microscope to look at real microfossils, and help you explore the groups we have covered a bit more. It'll also be pretty relaxed, and hopefully fun! These are real samples, which are rarely perfect (or in some cases not even that good), so a key thing I suspect we'll find is that identifying microfossils in real sediments is hard, and we're going to have to live with uncertainty.

The exercise part 1: We have a series of slides of prepared microfossils for you at the front of the room, which are relatively well preserved examples of microfossils. You can study these using the binocular microscope. Using your notes from the lecture, and the material at the front of this handout, there are a series of questions to help you think about the fossil you can use to guide your note taking available as a separate sheet at the front.

The exercise part 2: In addition to these we have a few bags of random sediment from a range of time periods at the front of the room. If you fancy, you can pour a small amount of one of these into a petri dish, and then use the paintbrushes to sift through it and see what you find. Please be careful with this material – don't spill any, and once you are done, upend it onto a creased sheet of paper and return it to *the bag from which you got it*. It would be problematic if these were mixed up. Happy microfossil hunting!

Below is an overview of the material covered in the lecture that will help you find the answers you need – do bear in mind that the groups we have covered are not exhaustive, and you may well find other microfossils in this practical!

Foraminifera



Calcite tests of selected benthic (left) and planktonic (right) foraminifera – Source: Pearson, P.N., 2012. Oxygen isotopes in foraminifera: Overview and historical review. Paleontological Society Papers, 18, pp.1-38.

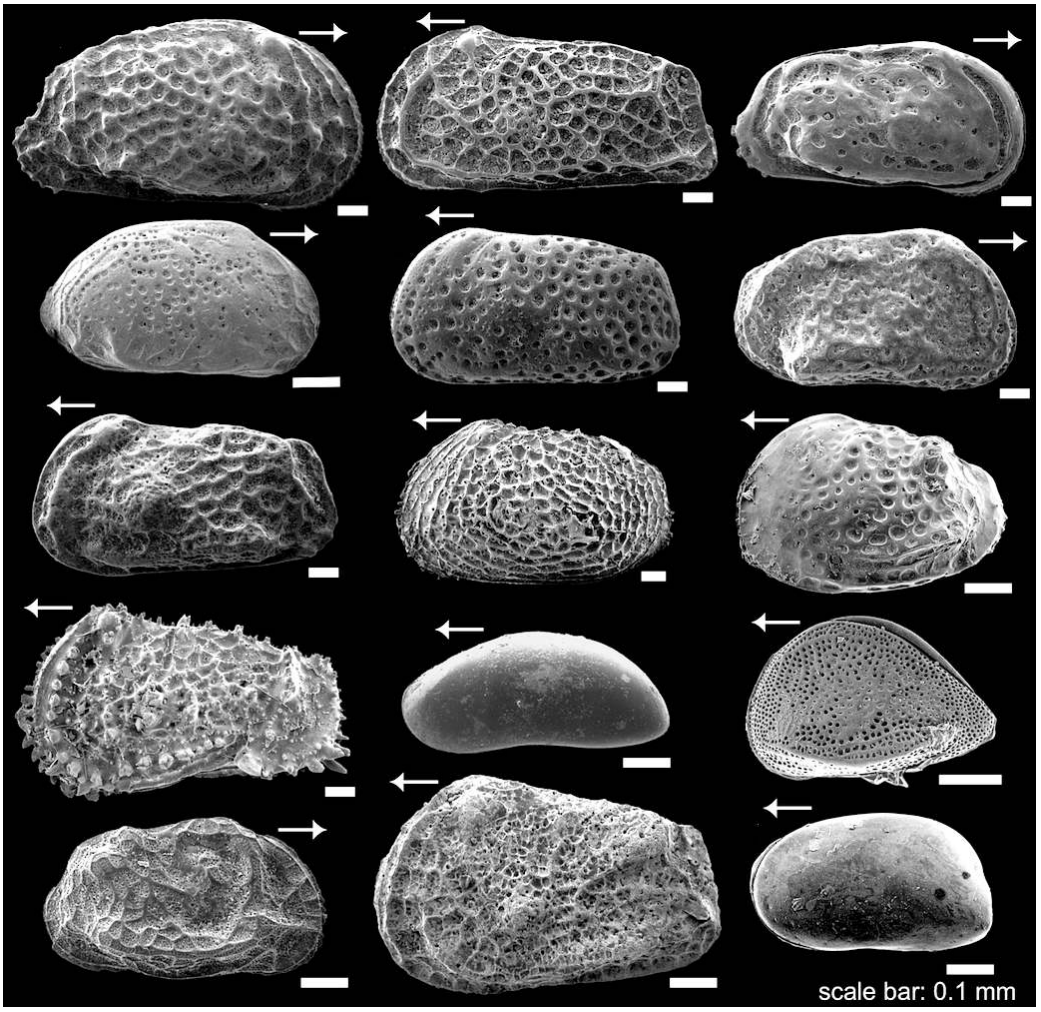
Singled celled organisms, which typically produce a test (shell) with one or multiple chambers – this often survives in the fossil record as a microfossil. Amoeboid protists, and a true grouping of organisms, in evolutionary terms.

Abundant in a variety of Phanerozoic sedimentary rocks. Age Range: Lower Cambrian to Recent. Most foraminifera have a mineralized shell (=tests) comprising chambers, or interconnected through holes (=foramina). Tests can be:

- Calcareous: randomly oriented crystals of high-Mg calcite.
- Calcareous hyaline: highly oriented calcite crystals.
- Agglutinated: fragments of extraneous material (siliciclastic/calcareous) bound together by a variety of cements.

Organisms may be motile (may have pseudopodia). Size: typically 0.05-0.5mm, but can be up to 20mm. Chamber arrangement and aperture shape are the key to identification. Earliest forms of forams are thought to be benthic. Benthic foraminifera are today found in marshes, brackish environments, carbonate platforms, reefs and back reefs, continental shelves and are also open marine. Whilst most forms are benthic, planktonic also exist (and have done since the mid-Jurassic). Foraminifera are found in waters from tropical, warm subtropical, and cool subtropical, to subpolar and polar. Invaluable in palaeoenvironmental reconstructions because each species is quite particular on where they live; for example, some only live in clean freshwater, others only live in the turbid waters near the mouths of rivers. Can also be used for studies in palaeobathymetry and palaeotemperature. Preserved in a variety of rock types, and preparation techniques used depend on the rock type. Hard rocks (e.g. some limestones) are best thin sectioned as in normal petrological studies. For loosely consolidated sediments, these can be broken down using water and detergent or up to 30% hydrogen peroxide, and the material is then sieved. Dried fossil samples can be picked from any remaining sediment using a fine brush and a reflected light, binocular microscope.

Ostracods



SEM images of fifteen modern ostracod species (Podocopida) from the coast of southern Okhotsk Sea. Source: Ozawa, H., 2013. The history of sexual dimorphism in Ostracoda (Arthropoda, Crustacea) since the Palaeozoic. Sexual Dimorphism. InTech Open Access Company, Rijeka, pp.51-80.

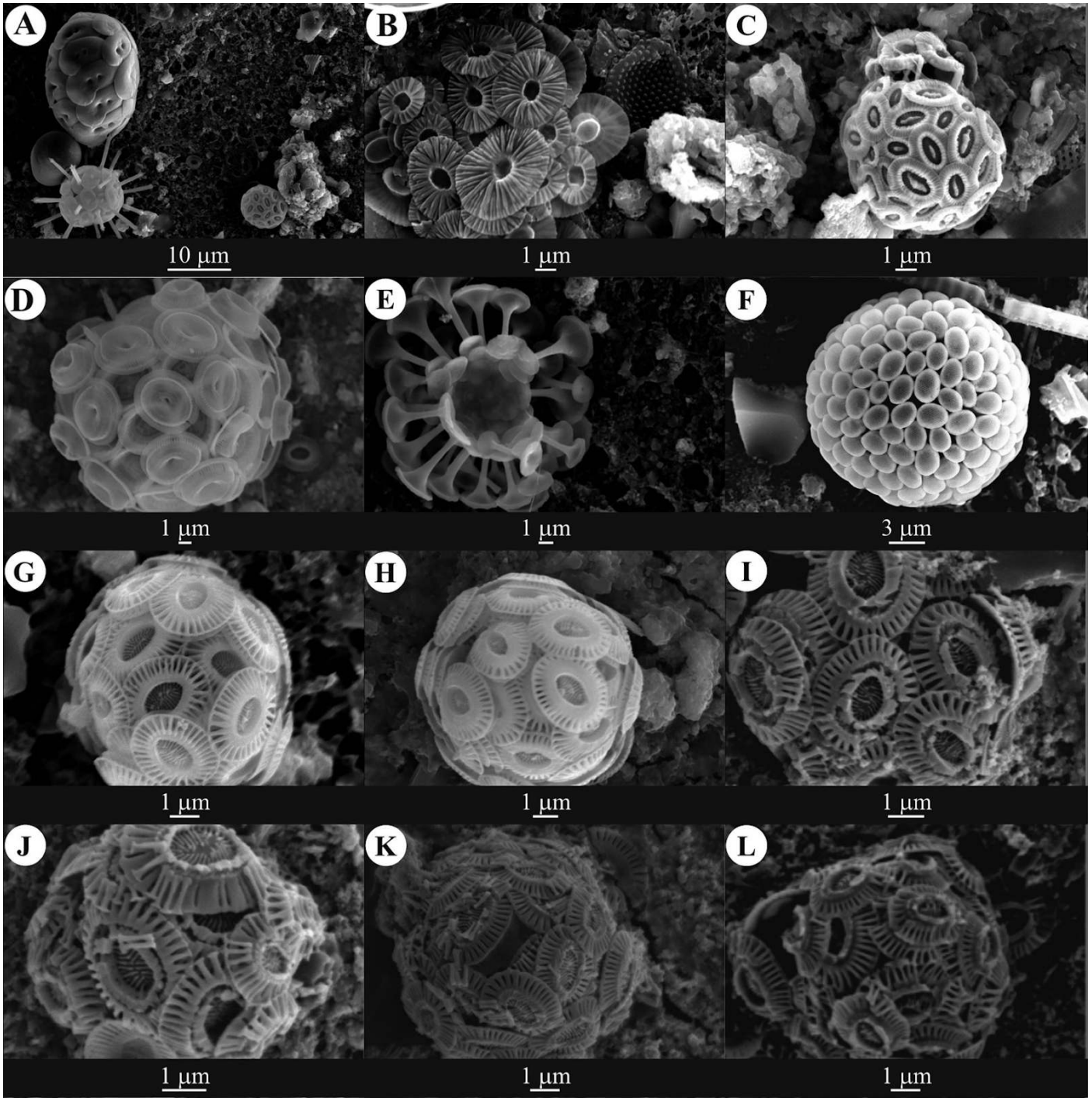
Crustacean arthropods whose carapace initially forms as chitin and then becomes calcified, and is preserved as a microfossil. As with foraminifera, a true grouping of organisms, in evolutionary terms – these are animals. Age range: Cambrian to Recent. Moult periodically. Body comprises a head and a trunk and is suspended within carapace by muscles. Bivalved carapace, hinges along the dorsal margin, and may have pore canals or sensory setae. 7 pairs of appendages, specialized as: sensory organs; for capturing and processing food; and for locomotion. As this is an animal it has digestive system, genitalia, nervous system, a median eye or a pair of lateral eyes. Sexual dimorphism is common – males commonly have a greater length to height ratio than females. Females may have a brood pouch. Size: most ~1mm but can be up to 10mm.

Most ostracods are benthic, swimming, crawling or burrowing at the sediment water interface (high organics); rare planktonic forms. First forms were marine, with freshwater forms from the Carboniferous. Geographic range from

the equator to the poles. Abundant and widespread in aquatic environments. Ostracod ecology can be inferred from the carapace shape and structure, and as a result,ostracods are useful palaeoenvironmental indicators.

Preparation of ostracods is similar to foraminifera – a rock can be fragmented and washed with hydrogen peroxide prior to sieving to isolate the fossils. Multiple washes can be used to break down some material, although this should be conducted carefully to avoid breakage of the specimens. One the samples have been dried, they can be picked from any remaining fragments with a fine brush under a reflected light, binocular microscope.

Calcareous nannoplankton



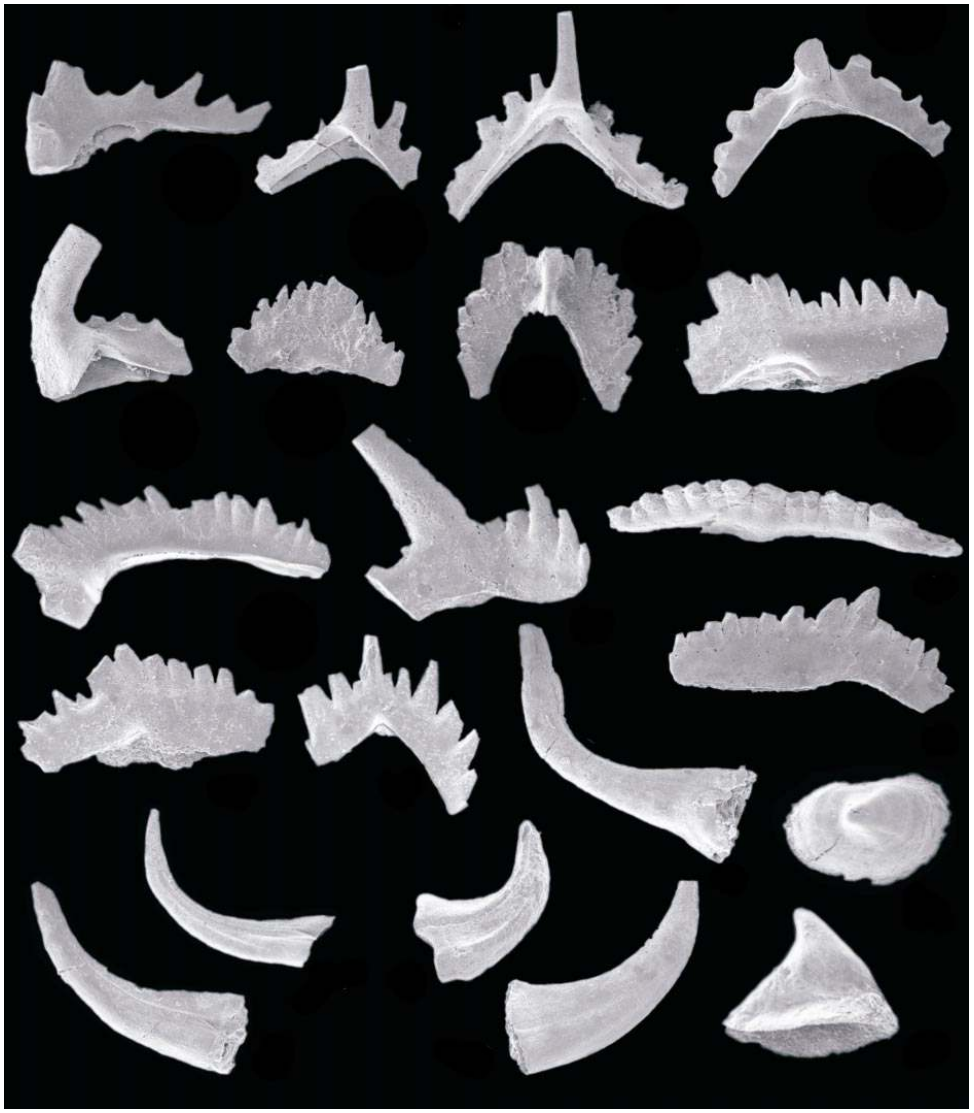
Scanning electron microscope images of coccolithophores collected in Levante Bay, November 2010. Source: Ziveri, P., Passaro, M., Incarbona, A., Milazzo, M., Rodolfo-Metalpa, R. and Hall-Spencer, J.M., 2014. Decline in coccolithophore diversity and impact on coccolith morphogenesis along a natural CO2 gradient. The Biological Bulletin, 226(3), pp.282-290.

Nannofossil formed from low-Mg calcite. Unicellular phytoplankton (a eukaryotic autotrophs that contain chloroplasts and can photosynthesize but may also be heterotrophic) – thus not a true group, but most common forms, coccolithophores, are – these are single-celled algae (division Haptophyta). Age Range: Late Triassic (Carnian)-Recent. Coccolithophores are organisms with an exterior layer called the coccosphere, probably for protection. This is made of coccoliths: calcareous structures formed by coccolithophore. Each is a disc-like plate secreted by the organism. An organic coating holds these together to form the coccosphere. Size: Generally less than 30 microns across and usually between 5-10 microns (individual coccoliths).

These are one of the major forms of phytoplankton found in the oceans today, and have a widespread distribution throughout the photic zone. They are found in almost all marine habitats, but are exclusively planktic marine organisms. Note that they may not be preserved in deep water sediments due to the CCD. Today there are greater concentrations in zones of high organic productivity where more nutrients are available due to upwelling – north and south of 45 degrees.

Calcareous nannofossils may be extracted from rock by simply doing a standard smear on a glass slide or through taking a scraping from the sample, putting it into a solution and placing it into a centrifuge. This results in a condensed pellet which can then be smeared onto a slide. The fossil can then be examined under a standard petrographic microscope or SEM.

Conodonts



Scanning electron microscope photomicrographs of conodonts from the Kenny Formation. Source: Orchard, M.J., 2010. Triassic conodonts and their role in stage boundary definition. Geological Society, London, Special Publications, 334(1), pp.139-161.

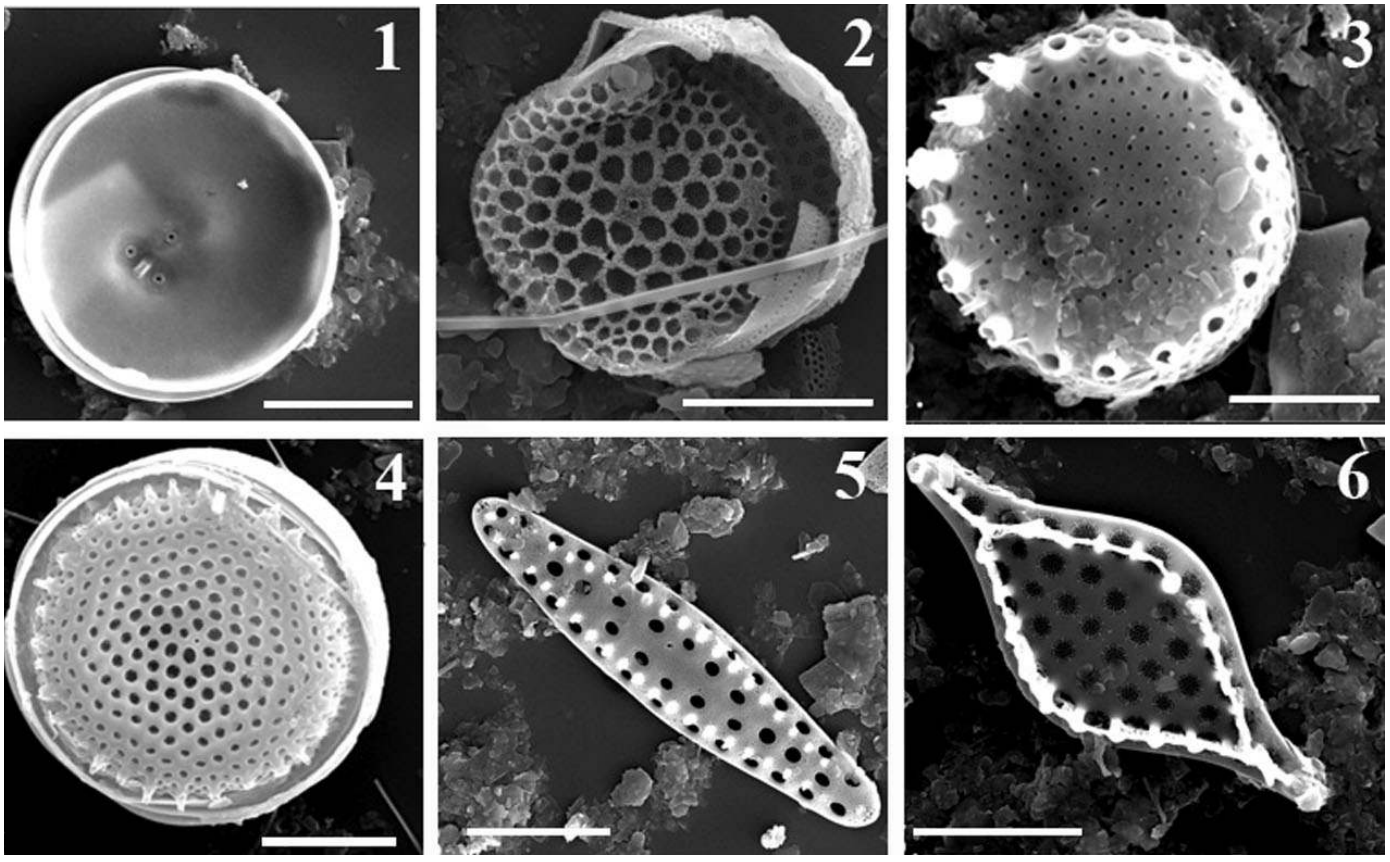
Phosphatic tooth-like microfossils made of calcium carbonate fluorapatite with additional organic matter, which are found as microfossils in (mostly) Palaeozoic rocks. Feeding apparatus of an extinct early vertebrate (=a conodontophorid) – a soft-bodied, bilaterally symmetrical organism. Age range: Precambrian to Triassic. Elements

classified into several broad types: simple cones, blade-type, bar-type, and platform. Note also that an anatomical notation scheme exists, naming elements based on their orientation with reference body axes (the P_n –S_n scheme). Relatively common in Palaeozoic rocks, where other microfossils are rarer, hence primary microfossils for working on Ordovician to Permo-Triassic strata. Colour changes of the elements can be related to the maximum thermal alteration of the sedimentary rocks. Size range: Individual elements are commonly 0.25–2 mm.

Generally found in marine to near-shore deposits, and common in black shales associated with graptolites, radiolarians, brachiopods, and fish remains. Beyond this little is known of the palaeoecology of the group: this is known only indirectly from the associated fauna and facies. It is likely that the group were entirely marine, and the fact that they are found in a wide range of different facies suggests they may have been nektonic or pelagic.

Chemical preparation techniques used to extract conodonts can use acetic, formic, or monochloroacetic acid, especially for carbonate host rocks. Furthermore, the elements can be prepared through mechanical approaches, such as sieving. Cleaned specimens can be viewed using a reflected light microscope (stereo/stereoscopic or dissecting microscope). They can also be observed in thin sections and be observed under SEM.

Diatoms

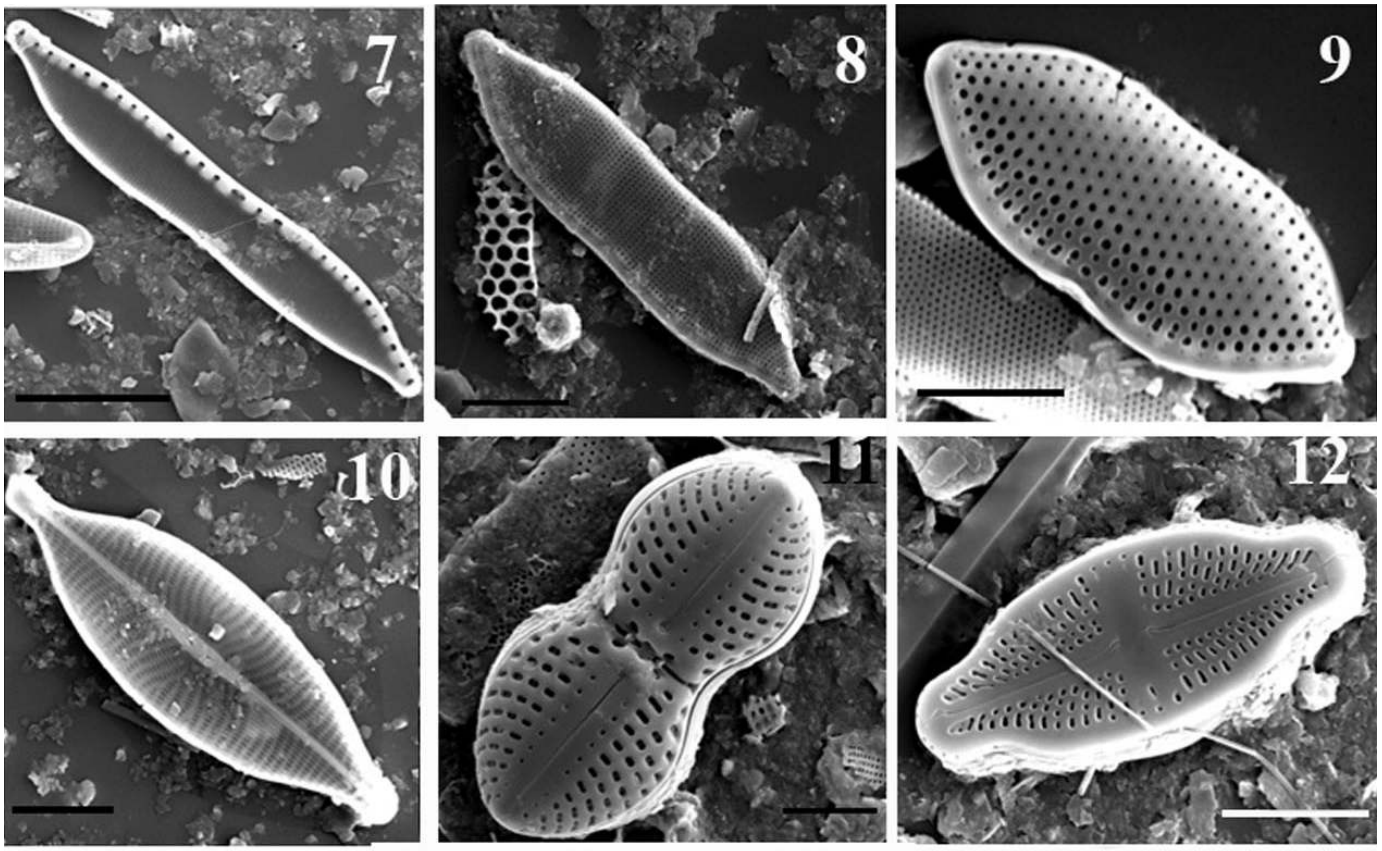


Diatoms from the Savannah River mud samples (Scanning Electron micrographs) – Continued overleaf. Source: Manoylov, K.M. and Dominy Jr, J.N., 2013. Changes in Epipelic Diatom Diversity from the Savannah River Estuary.

Group of algae that first appeared in the Jurassic. Predominantly unicellular autotrophs, but some multicellular forms, characterised by large green-brown chloroplasts. Age Range: Jurassic – Recent. Occur as individuals or loosely integrated colonies. Cells are enclosed in an external siliceous skeleton, the frustule, and when they die, the silica skeleton remains and sinks to the ocean floor. Two valves: larger called the epitheca, other the hypotheca. Two major groups: Centrales (1–4 above) which have round valves with pores radiating in concentric rows form the valve centre and Pennales (5–12), with elliptical valves with pores arranged in pairs. Size range: 20–200 microns diameter/length.

Diatoms as a group are found almost anywhere moist including: soils, freshwater, and marine environments. They are abundant in high latitudes and in low latitude upwelling zones, and both planktonic and benthic forms are known. Centrales are typically found in marine environments, whereas pennales are typical of freshwater environments.

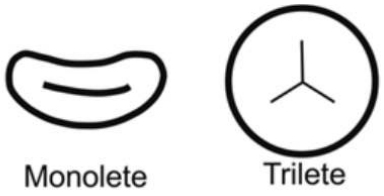
They only live in the photic zone but when they die can sink through even deep waters to accumulate as diatomaceous ooze or diatomite (sediments composed almost entirely of diatom frustules). They are particularly useful for biostrat in high latitude sediments where calcareous microfossils are often poorly preserved, sparse or of low density.



Preparation of diatoms is straightforward, and they can be viewed with the aid of a light microscope. Often samples can be scraped and then smeared onto slides, allowing examination. If further preparation is required, organics (which can obscure detail) are sometimes removed with hydrogen peroxide, and HCl can be used to dissolve calcium carbonate. Samples can then be rinsed with water, and the sample placed onto a slide. Observation normally uses light microscopes with a high objective, to allow for details to be seen considering the fossils' small size. SEM is also useful.

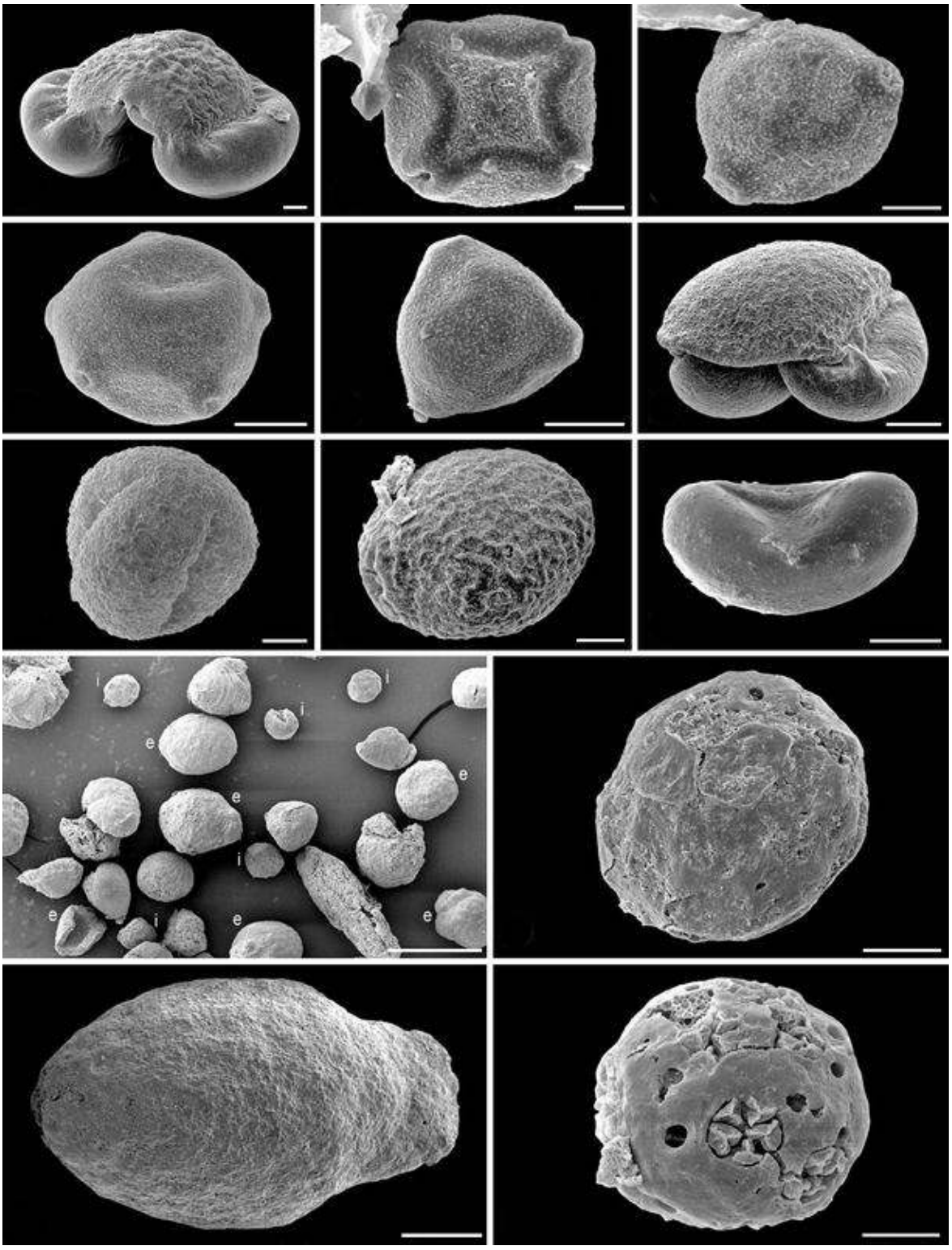
Pollen and Spores

Spores and pollen are produced in the life cycles of different types of plant. Extremely resistant organic structures, which are associated with the reproduction of these eukaryotes. Age Range: Silurian – Recent (cryptospores from mid Ordovician). Typically associated with higher plant groups, including gymnosperms (from latest Devonian) and angiosperms (from Early Cretaceous). Spores (produced by seedless plants) are found in forms known as Monolete and Trilete:



Megaspores several micrometres to millimetres (giant horsetails and club mosses). Microspores < 200μm diameter. Pollen is produced by seed plants and is extremely diverse in form. May have multiple pores and/or apertures (colpi). Covered with exine: the outer layer of the pollen is extremely tough and resistant. Pollen size 2-150 μm diameter. Spores and pollen are generally widespread, and because they can be carried by wind or water across both non-marine and marine environments are found in a wide variety of rocks. Their study is very important for the Upper

Palaeozoic, particularly Carboniferous Coal Measures rocks and associated biostratigraphy and extraction activities. Also regularly used to show thermal maturation of a rock and for palaeoenvironmental analysis.



Scanning electron micrographs of palynomorphs and sporangia from the Eemian of the Hooge Broek sand/gravel pit. Source: van der Ham, R.W., Kuijper, W.J., Kortselius, M.J.H., van der Burgh, J., Stone, G.N. and Brewer, J.G., 2008. Plant remains from the Kreftenheye formation (Eemian) at Raalte, The Netherlands. Vegetation History and Archaeobotany, 17(1), pp.127-144.

Organic-walled microfossils are the most labour-intensive and dangerous microfossils to prepare, because the acids used are very corrosive. Rocks including silts, clays, mudstones, sandstones, shales, and limestones are fragmented, and then treated with HCl to remove carbonates. In order to remove silicates the rock's residue is then bathed in HF. The acid must then be washed out, either through dilution or rinsing. The organic-walled components can then be studied using either light microscopes commonly (40x or greater objective is required for the small fossils) or using an SEM.

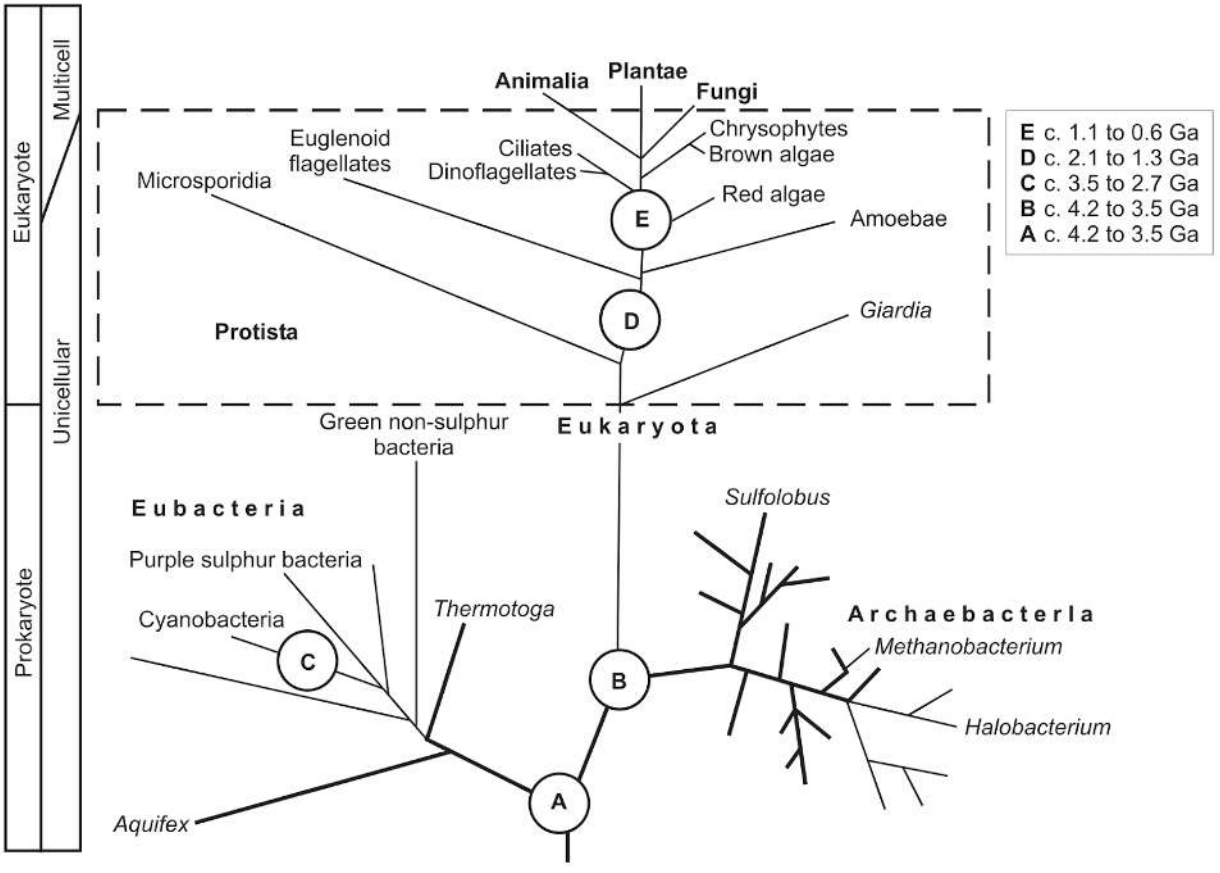
3D Models

There are also 3D models of all major groups of microfossils available on the website associated with this week's lecture:

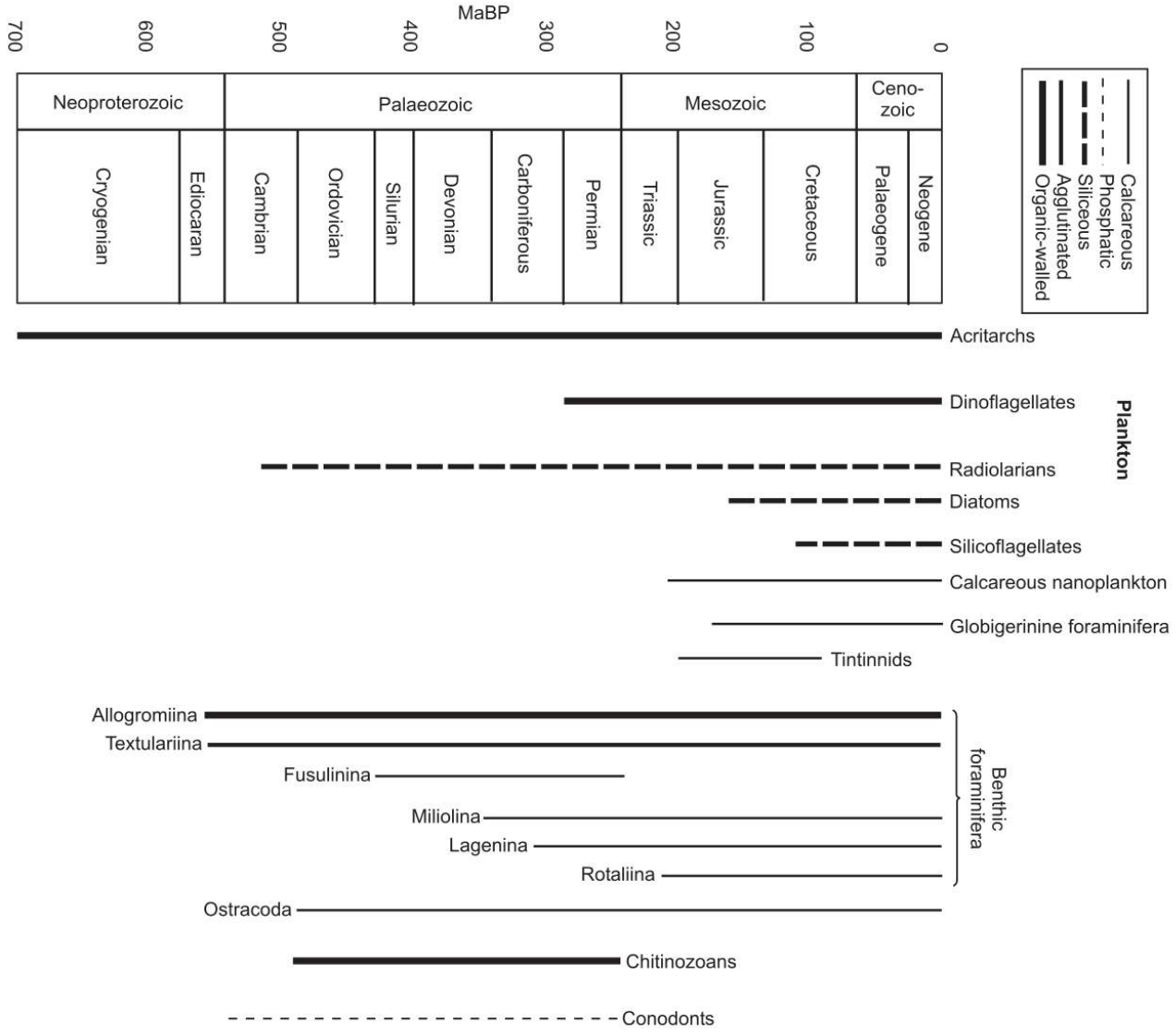
Link *Microfossils*

Diagrams

Included below are a couple of useful diagrams, which will put some of this into context:



The tree of life. Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.



Age ranges of important microfossil groups. Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.